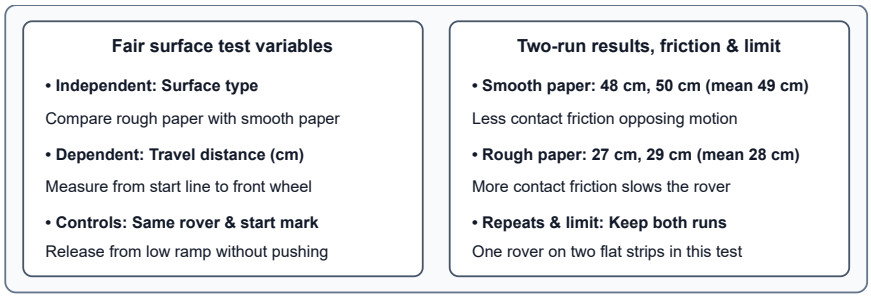
Reference sheets

Reference A · Fair surface test variables and planning rules

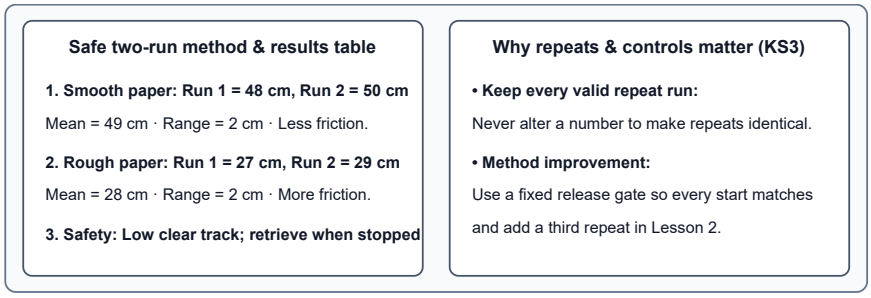


Fair surface test diagram. It shows independent, dependent, and control variables with two-run cm results.

Reference only. Do not write on this sheet. Check each variable role when building your surface plan.

Variable role.	What we do in our test.	Key Stage 3 science reason.	What happens if not controlled.
1. Independent variable (Change).	Change the surface type: rough paper vs smooth paper.	Tests how surface roughness changes contact friction.	If we do not change the surface, we cannot compare friction.
2. Control variables (Keep the same).	Keep the same rover, start mark, and no-push ramp release.	Ensures only surface friction causes any distance change.	A harder push or steeper ramp confounds the result.
3. Dependent variable (Measure).	Measure travel distance in centimetres (cm) from start line to front wheel.	Gives a clear number and unit for every run.	Without cm units, distances cannot be compared fairly.
4. Repeats and test limit.	Complete two repeat runs per surface and keep every valid number.	Shows whether the pattern holds across runs in this test.	One run alone cannot show if results vary.

Reference B · Two-run surface table and safe method



Two-run surface table diagram. It shows smooth and rough cm runs, safety rules, and test limits.

Surface condition.	Run 1 (cm).	Run 2 (cm).	Mean and Key Stage 3 explanation.
Smooth paper.	48 cm.	50 cm.	Mean = 49 cm (range 2 cm). Less contact friction opposes the rover.
Rough paper.	27 cm.	29 cm.	Mean = 28 cm (range 2 cm). More contact friction slows the rover.
Safety and method upgrade.	Low clear track.	Retrieve when stopped.	Keep hands and faces clear. Use a release gate to keep starts identical.

Made by Matt

Tier pack · Independent

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use the arrival diagram. Name the variable we change and the variable we measure.

Help. Check Box 1 for what changes and Box 3 for what is measured.

2. Examine Box 2 of the arrival diagram. Name two control variables we keep constant.

Help. List two factors that must not change between trials.

3. Why must we release the rover from the mark without pushing it by hand?

Help. Think about keeping the starting force identical every time.

4. Examine Box 3. Smooth runs are 48 and 50 cm. Rough runs are 27 and 29 cm. Which surface gave longer runs?

Help. Compare the centimetre numbers on both surfaces.

You do tasks

1. Use Reference A. State the independent variable and the dependent variable with units. Name three control variables for the plan. **Staff I ☐ S ☐ C ☐**

Help. Check the three variable rows in Reference A.

2. Use Reference B. Compare the two-run results (Rough: 27, 29 cm; Smooth: 48, 50 cm). Explain the difference using contact friction. **Staff I ☐ S ☐ C ☐**

Help. Calculate the mean for each surface and link to friction.

3. Use Reference A and Reference B. Why must we keep both repeat numbers when they differ slightly? State one way to improve the launch. **Staff I ☐ S ☐ C ☐**

Help. Check the Repeat and Limit row on Reference A.

Optional participation-loop prompt. Share your test plan and result with an adult. Hear one feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. State the independent variable, dependent variable, and two controls in our surface plan.

2. Why did the rover roll farther on smooth paper (48, 50 cm) than rough paper (27, 29 cm)?

3. What should you do if two repeat runs on the same surface are 2 cm apart?

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Independent

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt